

Static Grating Orientation Decoding

Allen Brain Observatory visual areas: VISP, VISI and VISpm

1. Objective

The aim of this analysis was to determine whether population neural activity in visual cortex could decode static-grating orientation. **Two decoding tasks:** (1) binary discrimination between 0° and 90°, and (2) six-class discrimination among 0°, 30°, 60°, 90°, 120° and 150°.

2. Decoding model and features

Neural features were population spike counts measured during the first 250 ms after stimulus onset. The classifier was logistic regression, implemented in a scikit-learn pipeline with StandardScaler followed by LogisticRegression. Performance was evaluated using stratified 10-fold cross-validation with shuffled folds and a fixed random seed. Area-specific analyses were performed for VISP, VISI and VISpm. To compare areas more fairly, neuron count was also matched to 40 neurons per area using repeated random subsampling (20 repeats).

3. Binary orientation decoding: 0° vs 90°

Population spike counts strongly separated the two orientations. Combining neurons from VISP, VISI and VISpm produced a mean decoding accuracy of **96.45%** (chance level 50%). This indicates that orientation information was highly separable at the population level for this stimulus condition.

3.1 Each area using all available neurons

Area	Neurons	Mean accuracy	CV standard deviation
VISP	110	88.07%	3.42%
VISI	40	90.60%	5.72%
VISpm	66	94.40%	3.39%

Using all available neurons, VISpm achieved the highest area-specific decoding accuracy. However, this comparison is affected by unequal population sizes across areas.

3.2 Fair comparison using 40 neurons per area

Area	Matched-neuron accuracy	Variation across sampled populations
VISP	82.88%	2.98%
VISI	90.60%	<1%
VISpm	91.46%	3.64%

After matching the population size, VISI and VISpm remained highly accurate, at approximately 90.6% and 91.5%, respectively, while VISP decreased to approximately 82.9%. The result suggests stronger population-level separability in VISI and VISpm than VISP for this session.

4. Six-orientation decoding

The six-class task was substantially more difficult than binary decoding, but performance remained well above chance. Using neurons from all three areas, the mean accuracy was **66.64%**, compared with a chance level of **16.67%**. The standard deviation across cross-validation folds was 3.36%.

4.1 Each area using all available neurons

Area	Neurons	Accuracy
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VISp	110	60.64%
VISI	40	53.85%
VISpm	66	63.04%

Using all available neurons, VISpm showed the highest six-orientation accuracy, followed by VISp and VISI. Because neuron counts differed substantially, the matched-neuron analysis provides the fairer comparison.

4.2 Fair comparison using 40 neurons per area

Area	Matched-neuron accuracy
VISp	49.10%
VISI	53.85%
VISpm	56.52%

With exactly 40 neurons per area, VISpm achieved the best performance, followed by VISI and VISp. All three areas remained substantially above the 16.67% chance level, demonstrating that each area contained information about stimulus orientation.

5. Summary

Binary 0° versus 90° decoding was particularly strong, reaching 96.45% when neurons across the three visual areas were combined. Six-orientation decoding was more challenging but remained clearly above chance at 66.64%. After controlling for neuron count, VISpm showed the strongest performance in both tasks, with VISI also performing strongly. These findings support the presence of robust orientation-related population information across VISp, VISI and VISpm in this recording session.

Model note: Although this analysis was previously labelled “linear regression,” the notebook uses logistic regression, which is a linear classification model appropriate for categorical orientation decoding.